

Research paper

Enhancing thermal efficiency of double-pass solar air collectors: A comparative study on the role of V-angled perforated fins

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ABSTRACT

This study investigates the enhancement of thermal efficiency in double-pass solar air collectors (DPSACs) by incorporating V-angled perforated fins. It compares the performance of a finned model (FDPSAC) with a conventional smooth model, aiming to optimize thermal efficiency through improved heat transfer and air distribution. The experiments were conducted under three airflow rates: 0.00864 kg/s (Case I), 0.01143 kg/s (Case II), and 0.01317 kg/s (Case III). The findings indicated that the FDPSAC significantly outperformed the DPSAC, achieving higher outlet temperatures and demonstrating superior heat gain. The daily efficiency for the FDPSAC was 51.6 %, 54.7 %, and 59.2 %, exceeding the DPSAC, which were 49.4 %, 50.5 %, and 53.9 % for cases I, II, and III, respectively. The daily collector efficiency of the FDPSAC improved by approximately 4.45 %, 8.32 %, and 9.83 % relative to the DPSAC for the same case order. The results also showed that the FDPSAC maintained lower temperatures of the absorber plate, suggesting more effective heat dissipation and overall system efficiency. This study highlights the critical role of innovative fin design and optimal airflow management in enhancing solar collector performance. These improvements highlight the potential of the proposed fins to enhance solar air collector efficiency, providing key insights for optimizing thermal systems in sustainable energy.

1. Introduction

The current global energy crisis is defined by a sharp increase in energy use and CO₂ emissions, especially from heating, leading to very high levels of emissions [Ammar et al. \(2020\)](#). This scenario highlights the need for sustainable energy [Almadhhachi et al. \(2022\)](#). In this context, sustainable energy has seen a significant focus on solar power, notably for its widespread accessibility and abundance [Ali et al. \(2023\)](#). This interest has led to much research into solar collector technology, mainly for using solar energy for heat. It emphasises solar energy's role as an effective and readily available solution for energy and environmental challenges [Alshibil et al. \(2023\)](#).

Solar air collector (SAC) collects and transforms solar energy into thermal energy, which is ideal for low-temperature uses such as heating and drying process [Hedau and Singal \(2023\)](#). A flat plate SAC typically features a channel with a glass top surface that allows sunlight to penetrate and hit the black-painted metal surface that absorbs heat. The

metal surface, serving as the channel's bottom, is surrounded by insulation on the sides and below the absorber to minimize heat losses. As air moves through the channel, it captures heat from the metal surface, making the FPSAC useful for thermal applications. The design of these air collectors can vary, and they may include multiple air channels. Depending on the intended application, there are single pass solar air collectors (SPSAC), double pass solar air collectors (DPSAC), and triple pass solar air collectors (TPSAC), each with increasing numbers of air channels to improve heat transfer efficiency.

Nonetheless, SACs with smooth surfaces often suffer lower performance because of the limited transfer of heat across the fluid and the absorbing surface. As the air passes over a heated absorbing surface, a thin layer forms between them, slowing heat transfer. This effect, known as a laminar boundary layer, can be reduced by causing the airflow to become turbulent. For that, various enhancement methods like grooves, baffles, pins, and fins are employed to enhance heat transfer, although they can increase friction and pumping needs [Hamad et al. \(2023\)](#); [Machi et al. \(2024\)](#), (2022); [Vengadesan et al. \(2024\)](#).

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Nomenclatures			
A_p	the projected area of the collector (m^2)	Δp_{or}	the pressure drops across the orifice plate (pa).
A_{or}		Δp	the pressure drops across the collector pass (pa)
A_{ch}	the orifice plate area (m^2)the air channel area (m^2)	Q_u	useful heat gain (W)
T	temperature (C)	η	collector thermal efficiency
I	solar radiation on inclined surface (W/m^2)	β	the ratio between the diameter of the orifice and that of the duct.
C_p	air specific heat ($\text{J}/\text{kg}\cdot\text{C}$)	ρ_o	the density of the air at the outlet temperature.
$\dot{m}$	air mass flowrate (kg/s)	ε	
h_r	the radiative heat transfer coefficient ($\text{W}/\text{m}^2 \text{ K}$)	μ	the emissivitydynamic viscosity of the air ($\text{kg}/\text{m}\cdot\text{s}$)
h	the convective heat transfer coefficient ($\text{W}/\text{m}^2 \text{ K}$)	Subscripts	
U	the heat lost coefficient ($\text{W}/\text{m}^2 \text{ K}$)	i	inlet
k	thermal conductivity($\text{W}/\text{m K}$)	o	outlet
x_b	insulation thickness (m)	p	plate
V_w	surrounding wind velocity (m/s)	g	glass
L	length of the absorber (m)	s	sky
W	width of the absorber (m)	f	the arithmetic means of air.1
B	hight of the air channel (m)	$1,2$	1st and 2nd air stream
H	fin hight (m)	b	Bottom insulation
D_h	the hydraulic diameter of the flowing channel (m)	fin	fins
Re	Renolds number	t	total top
Nu	Nusselt number	w	wind
C_d	discharge coefficient.	or	orifice

Zhu et al. (2017) introduced an innovative SAC featuring micro-heat pipe arrays (MHPA) to improve thermal efficiency. They conducted multiple experiments to assess the heat transfer and friction factor properties of SAC. Numerical simulations were also developed to model the collector's performance under quasi-steady conditions. The results indicated an average efficiency of around 69 % at 290 m^3/h volumetric flow, while friction coefficient ranged between 0.05 and 0.18. They claimed that the proposed design, improved absorption efficiency, enhanced heat transfer, and reduced airflow resistance.

Elakroun et al. (2024) numerically analysed using CFD the thermal performance of DPSAC with three different rectangular fin arrangements: parallel, vertical, and opposed. The research revealed that SAC with parallel fin notably enhanced heat transfer, resulting in an outlet temperature of around 44.6C at MFR of 0.011 kg/s .

Hassan et al. (2021) analysed in depth the performance of a DPSAC utilizing smooth, V-corrugated, and corrugated-perforated finned absorbers, evaluated across various air mass flow ratios. The ratios included single pass (SP), one-third double pass (1/3 DP), two-thirds double pass (2/3 DP), and full double pass (DP). Their experiments identified the 2/3 DP corrugated-perforated absorber configuration causing the highest enhancement in both energetic and exergetic efficiencies. This setup reached the highest temperature of 44.6C at the outlet port and demonstrated an average daily thermal efficiency of 71.85 %. Moreover, the average daily exergy efficiency for this configuration was 0.975 %.

Vengadesan et al. (2024) conducted an experimental study on two DPSACs. The first collector featured continuous tubular elements combined with V-corrugated fins, while the second utilized discrete tubular elements with V-corrugated fins. The proposed continuous finned absorber increased heat transfer coefficients by 87.2 % and discrete fins by 59.3 %. Nusselt numbers (Nu) rose by 100 % for discrete fins and 44.6 % for continuous fins. Peak energy efficiency reached 82.5 % for discrete fins and 79 % for continuous fins. The thermo-hydraulic performance factors were 2.01 for discrete fins and 1.6 for continuous fins. This configuration reduced heat loss by 10.7–23.3 % in discrete fin structures and 6.7–19.5 % in continuous fin structures.

Sharma and Debbarma (2024) experimented the improvement of reverse flow SAC using a perforated absorber combined with delta-wing turbulators. They noted significant enhancements in both heat transfer

and thermohydraulic performance, especially with the use of inline delta wing configurations. They noted significant enhancements in heat transfer and thermohydraulic performance, especially with inline delta wing configurations. The study found that an inline deltawing turbulator with perforations resulted in a 6.6-fold increase in the Nusselt number and a 9.4-fold rise in the friction factor when set at a 75° attack angle. The overall thermohydraulic performance improved by 3.06 times compared to smooth ducts. Ozgen et al. (2009) explored three absorbing surface arrangements in parallel flow DPSAC. They discovered that incorporating aluminium cans in the collector will increase the heat transfer area, producing turbulence, hence increasing convection heat coefficient. A staggered cans arrangement attained maximum thermal efficiency at 0.05 kg/s MFR exceeding 70 %.

In their study, Abdullah et al. (2018) evaluated a counterflow DPSAC with aluminium cans working as a fin and compared it to aSPSAC. They tested various fin configurations in the DPSAC: flat, aligned, staggered, and staggered with air guidance. The DPSAC with flat configuration showed a 22 % increase in Nu compared to the SPSAC. The aligned and staggered arrangements improved the Nu by 58–68 % at 0.05 kg/s MFR. Moreover, DPSAC with staggered cans and air guidance achieved up to 68 % daily efficiency, exceeding the flat DPSAC and the standard SPSAC. Likewise, Al-Damook et al. (2019) studied the efficiency of DPSACs using recycled aluminium cans as turbulators. They used COMSOL Multiphysics for CFD modelling of three designs: cocurrent, counter-current, and U-shape. Three U-shaped models were compared in outdoor tests: one without cans, one with cans in-line, and another with cans staggered. The U-shape model was more energy-efficient by 5.4 % and 6.5 % compared to the other two. Among the three models, the staggered cans model showed the highest efficiency at 60.2 %, better than the in-line (53.1 %) and plain models (49.4 %). Hussein et al. (2023) examined the thermal efficiency of DPSAC by replacing traditional flat plate absorbers with tubular ones oriented either perpendicular or parallel to the airflow. The parallel-oriented tubular absorber reached effective efficiency exceeding 80.5 % at 0.03 kg/s , outperforming the flat plate and perpendicular models by 4.2 % and 9.8 %, respectively. The useful energy peaked 292 W at the same MFR. The study highlighted that higher mass flow rates lead to increased energy gains and that absorber orientation significantly impacts thermal efficiency.

Raj et al. (2019) analysed the DPSAC performance by studying the

influences of fluid properties, channel depth, and flow rates. The research identified solar radiation and surrounding temperature being the primary factors influencing energy and exergy efficiency, with relative humidity exerting a negligible effect. Experimental findings indicated that thermal efficiencies ranged from 20 % to 41 %, while exergy efficiencies varied between 5.6 % and 18 % at 0.02 kg/s and 0.03 kg/s MFRs, respectively. Moreover, the air temperature remained consistently higher in the lower channel compared to the upper channel. The collector's performance was significantly affected by channel depth, ambient conditions, and variations in mass flow rate.

El-Sebaei et al. (2011) conducted a study to evaluate the performance of DPSACs featuring both smooth and V-corrugated surfaces. The research demonstrated that the V-corrugated DPSAC outperformed the smooth version by 11–14 %, with the highest thermo-hydraulic efficiencies observed at 0.02 kg/s MFR. Moreover, the maximum outlet temperature achieved for the V-corrugated DPSAC was 64.5 °C. Additionally, V-corrugated DPSAC had higher output power at 474 W and lower overall heat losses at 546 W compared to the smooth DPSAC.

Sudhakar and Cheralathan (2020) performed an experimental analysis on a DPSAC featuring pin-fins combined with a corrugated absorber plate to optimize its performance for drying agricultural products. The findings highlighted the importance of airflow in determining efficiency. At a flow rate of 0.035 kg/s, the proposed absorber achieved a 17.4 % increase in thermal efficiency compared to the conventional design. Moreover, the average outlet temperature increased by 12 %, with a maximum temperature difference of 14.8°C. The system attained a peak efficiency of 52 %, significantly surpassing traditional designs.

Karim et al. (2014) developed a computational model for a corrugated DPSAC using MATLAB, which showed a maximum deviation of 7 % compared to experimental data. The study identified solar radiation, MFR, and inlet air temperature, along with collector length, as key factors influencing efficiency. Moreover, the MFR and inlet temperature were found to be the most impactful. The research suggested that DPSAC systems provide superior efficiency compared to SPSAC systems. Kabeel et al. (2018) conducted experiments on a SPSAC modified with longitudinal fins and an enhanced entrance. They investigated the thermal performance impact of fin heights of 3, 5, and 8 cm alongside guide blades at the collector's entrance for better air distribution. The study found that, compared to a traditional SAC, the modified version with 8 cm fins achieved an approximate 57 % daily efficiency at 0.04 kg/s MFR. This modification outperformed the standard collector, which had an efficiency of 32 % in the same mass flow rate.

Salih et al. (2019) investigated a trapezoidal DPSAC during winter, comparing natural and forced convection. Rectangular aluminium fins affixed to either side of the absorber enhanced heat exchange. Results showed that natural convection achieved a higher efficiency of 73.5 %, while forced convection achieved 65 %. The highest temperature difference was 45.5°C and 17°C for natural and forced convection, respectively. The study concluded that natural convection is better for applications requiring higher temperature differences, such as drying, while forced convection is preferable for household heating applications. Moreover, Salih et al. (2021) extended their research to compare finned DPSAC with and without a porous medium placed above the absorbing surface, in both natural and forced system. The study found that the DPSAC with the porous medium significantly improved performance. The peak efficiencies for the porous one were 87 % under natural convection and 81 % under forced convection, compared to 82 % and 67 % for the DPSAC without porous medium. The results highlighted the substantial impact of the porous medium and air circulation type on heat transfer rates, making it more suitable for high-temperature applications like drying processes.

Singh (2020) enhanced a finned solar air collector's thermal efficiency through experimental and numerical methods, focusing on geometric and flow parameters. The findings reveal that the hybrid staggered fin configuration increases thermohydraulic performance by 3.5–3.8 % over other configurations. Additionally, the DPSAC achieved

79 % efficiency, surpassing the SPSAC by 13 %. Farzan and Zaim (2023) experimentally studied a DPSAC having perforated absorbers with steel wool as porous materials. Their findings showed that the porous DPSAC achieved higher exhaust temperatures of 58°C and 53°C at 0.012 kg/s and 0.024 kg/s MFR, respectively, compared to 54°C and 48°C for the non-porous DPSAC. The porous collector also exhibited a thermal efficiency improvement of 11.8 % and 15.3 % at the respective flow rates, with daily efficiencies reaching 49.1 % and 66.1 %, respectively.

Ramani et al. (2010) explored the thermohydraulic performance of DPSACs using theoretical and outdoor experiments with and without porous additions. Their research revealed that incorporating porous material into the DPSAC led to a 20–25 % efficiency improvement compared to the DPSAC without porous material and a 30–35 % enhancement over the SPSAC. This improvement was attributed to the increased surface area available for heat transfer. The double pass arrangement also reduced heat loss, making it ideal for high-efficiency applications like food drying.

Chang et al. (2024) presented a numerical assessment using CFD Fluent to optimize heat transfer efficiency in a SAC with longitudinal and transverse baffles. They found that transverse baffles increased air velocity and reduced vortex zones, leading to a 2.34–6.97 % greater efficiency than longitudinal baffles. Four baffles attained an efficiency of 54.32 % at 0.026 kg/s, while six baffles were optimal for higher rates, achieving efficiencies between 60.03 % and 64.39 %. The study concluded that transverse baffles significantly enhance SAC performance.

El-khawajah et al. (2011) conducted a study on DPSAC configured with 2, 4, and 6 fins, complemented by the integration of wire mesh for enhancing heat collection. They analysed the impact of MFR between 0.0121 and 0.042 kg/s on the collector exit temperature and efficiency. The study revealed that efficiency rises as the MFR and the number of fins increased, peaking at 85.9 % with six fins at 0.042 kg/s. The fins caused a significant temperature difference, highlighting the efficiency of varying fin configurations at different flow rates.

Chabane et al. (2020) experimentally studied the impact of rectangular baffles on heat transfer and pressure drop in a SPSAC. The objective of the study was to improve the thermal efficiency of SAC while keeping the pressure drop minimal. Different configurations featuring various numbers of baffles were tested. The configuration with 18 baffles achieved the highest thermal efficiency of 70 %, followed by configurations with 14 baffles at 65 %, ten at 65 %, six at 58 %, and three at 50 %. Additionally, positioning six baffles in the centre of the collector showed the best performance, balancing high thermal efficiency with manageable pressure drop.

Karwa (2024) conducted an experimental study to improve the efficiency of a SAC with an absorbing plate featuring longitudinal inclined fins forming multiple equilateral triangular airflow passages. The study found that using aluminium fins and a triangular passage width of 26.5 mm for a 2 m collector length enhanced thermal efficiency by up to 25.8 % compared to conventional designs. The research concluded that this configuration significantly increased heat transfer efficiency without requiring additional pumping power, offering a notable improvement over traditional SACs.

Singh and Kumar (2023) experimentally studied a SAC with frustum-shaped roughness elements. The study tested various configurations, finding that the highest thermal efficiency of 79 % was achieved at a relative roughness pitch of 12 and 0.0486 kg/s MFR. The roughened plates significantly improved heat transfer compared to smooth ones, although increased roughness also raised the friction factor, potentially increasing energy requirements. The research emphasizes the need to optimize roughness geometry for a balance between thermal efficiency and pressure drop.

In similar studies, researchers have examined ribs with various shapes, including circular, semicircular, and triangular. Initially, circular ribs were preferred due to their simple construction. Over time, their use has declined due to lower thermal-hydraulic performance parameter

(THPP). They struggle with heat transfer due to limited rib-top area. However, circular ribs are still used for arc-shaped designs because of their bending ease. V-shaped and inclined ribs, with or without gaps, are now preferred. Recent literature focuses on V-shaped ribs. These are often made in rectangular or square sections for better heat transfer in SACs.

Gabhane and Kanase-Patil (2017) conducted experimental research on a SAC with double-flow design that incorporated multiple C-shaped ribs as elements to artificially induce surface roughness. Their findings demonstrated that this configuration attained a maximum Nusselt number of 415 when the relative roughness pitch (P/e) was set at 24, and the roughness angle was maintained at 90° , coinciding with a Reynolds number of 15,000. The observed friction factor was 0.031, and the thermohydraulic performance parameter reached 3.48. This optimized arrangement markedly enhanced heat transfer efficiency relative to traditional single-flow setups. Hedau and Singal (2023) performed a numerical analysis on a DPSAC featuring semicircular tubes and perforated baffles. Their results revealed that the maximum Nusselt number achieved was 274.29, occurring at a Reynolds number of 19,000, while the highest friction factor was 0.6290 at a Reynolds number of 3000. The study also noted that increasing the tube amplitude ratio (a/H) from 0.2 to 0.5 elevated the Nusselt number from 65.64 to 242.72. Also, the friction factor rose with higher a/H ratios, peaking at 0.2155 when the Reynolds number was 3000. The researchers concluded that these configurations promote heat transfer enhancement but also increase friction, suggesting that these effects could be balanced through geometric modifications.

Singh et al. (2022) experimentally investigated the performance of a parallel flow DPSAC with perforated and continuous multi-V ribs. They found that perforated ribs significantly improved heat transfer and reduced friction compared to continuous ribs, with a maximum Nusselt number of 481.14 at Reynolds of 18000. The perforated ribs enhanced thermal performance, achieving a thermo-hydraulic performance parameter 2.97.

In their research, Ravi and Saini (2016) studied DPSAC with V-shaped and staggered ribs, examining relative staggered rib sizes between 1 and 2.5 and reporting significant improvements in thermal efficiency. They found that the roughened geometry significantly improved heat transfer and increased friction compared to smooth ducts, with a maximum Nusselt number of 440.47 at Reynolds of 20000. The roughened DPSAC exhibited enhanced thermal performance, achieving the highest increment in the friction factor of 3.2 times that of the smooth duct. Moreover, Ravi and Saini (2018) expanded their study to assess variations in rib pitch, sizes, and roughness. The study demonstrated that roughness increased the Nusselt number in the DPSAC up to 4.52 times compared to its smooth counterpart and 9.64 times compared to the SPSAC. The friction factor also rose by 3.13 and 8.53 times, respectively. Optimal roughness parameters included a relative rib size (r/e) of 3.5, a rib pitch (p/p) of 0.6, and a roughness width (W/w) of 7. These adjustments significantly boosted DPSAC performance.

Sharma et al. (2017) performed an experimental investigation on a DPSAH featuring V-shaped ribs for surface roughness. Their results indicated that the roughened DPSAC exhibited maximum Nusselt numbers and friction factors compared to smooth plates, with parameters being a relative roughness pitch of 10, relative roughness height of 0.044, and an angle of attack of 60° . The developed correlations for the Nusselt number and friction factor facilitate the prediction of heat transfer and pressure drop characteristics for such heaters. The results revealed the effectiveness of using V-shaped ribs in enhancing DPSAC performance.

The majority of research on employing V-shape ribs as roughened surfaces has focused on single-pass solar air collectors (SPSAC). Alam et al. (2014a) enhanced SPSAC heat transfer using V-shaped perforated blocks, achieving 33 % higher Nusselt numbers and reducing friction factors by 32 % versus solid blocks. Maximum Nu was at a blockage ratio of 0.8, with peak friction at e/H of 1.0 due to shear layer reattachment.

Nusselt numbers increased with open area ratios up to 20 %, and were lower for 5 % open area ratios due to expanding jet diameters. Alam et al. (2014b) further validated these findings, achieving maximum heat transfer and developing precise correlations for Nusselt numbers and friction factors, with deviations of 5.03 % and 4.99 %, respectively, demonstrating the effectiveness of V-shaped perforated blocks in enhancing SPSAC performance. Kumar et al. (2017) studied the heat transfer process in SACs using multi-V-down pattern perforated baffles to refine their geometrical parameters. Their study indicated that adjusting the baffle width, while keeping baffle height, pitch, hole position, and open area ratio constant, influenced thermal performance distinctly. Optimal heat transfer was observed at 5.0 relative baffle width ratio. They reported the highest Nusselt number ratio and friction factor ratio at this width, demonstrating the effectiveness of baffle adjustments in enhancing SAC performance.

Likewise, Jain and Lanjewar (2019) investigated the effect of V-ribs with symmetrical gaps and staggered configurations on an SPSAC absorber plate, focusing on the influence of the P/e ratio on performance. They discovered that Nusselt numbers and friction factors significantly improved with 2.30 and 3.13 compared to a smooth SPSAC, highlighting the peak in thermohydraulic performance. Furthermore, Deo et al. (2016) experimentally studied a SPSAC roughed by multi-gap V-down ribs combined with staggered ribs. Their result revealed that optimal enhancement attained by a relative roughness pitch of 12, a roughness height of 0.044, and 60° angle of attack. Their study showed that using broken V-down and staggered ribs improved the thermohydraulic performance parameter by 2.45 times. This emphasizes the effectiveness of creating gaps in rib roughness at specific geometrical conditions.

Jin et al. (2017) demonstrating that multiple V-shaped rib significantly outperform inline configurations in SACs, attributed to the enhanced generation of subsidiary vortices and flow redevelopment. They attained a peak thermohydraulic performance factor of 2.43 in SACs using staggered multiple V-shaped ribs, emphasizing the critical role of rib geometry in optimizing thermal efficiency.

Hans et al. (2010) achieved the most significant enhancements, with Nusselt numbers and friction factors increasing up to six and five times in SACs featuring multiple V-rib (W-shaped) configurations, underlining the importance of optimal roughness parameters. Collectively, these studies underscore the critical role of precise rib configurations and gap placements in maximizing the thermal and thermohydraulic performance SACs. Kumar et al. (2016) conducted experiments on solar air channels with broken multiple V-type baffles, examining how baffle relative width influences thermal performance. They discovered that a relative width of 5.0 maximized heat transfer efficiency, with broken V-type, baffles outperforming smoother designs due to enhanced heat transfer and decreased friction, making them an optimal option for solar air channel applications.

Kumar et al. (2024) recently used CFD Fluent to enhance thermal performance in a SPSAC with V-ribs and ring turbulators. They found that optimal thermal performance was achieved by setting specific ratios for W_D/W_{VR} (duct width to V-rib width), P_{VR}/H_{VR} (V-rib pitch to height), and H_{VR}/D_h (V-rib height to hydraulic diameter), with the most significant being a W_D/W_{VR} ratio of 5.0. This configuration achieved the highest thermohydraulic performance of 2.58 at a $Re = 16,000$. Additionally, it reached the highest Nu of 289 at a $Re = 20,000$, significantly improving heat transfer efficiency.

The above literature survey underscores the effectiveness of employing fins, ribs, and other roughening elements to enhance the performance of SAC. These modifications augment the heat transfer coefficient and rate by increasing the surface area and disrupting the boundary layer adjacent to the absorbing surface. Based on that, it is reasonable to hypothesize that integrating V-angled perforated fins into DPSACs may significantly enhance airflow turbulence, thereby boosting heat transfer efficiency. Notably, the shape and orientation of these fins are chosen to effectively double the heat transfer area compared to

standard V fins. Despite these theoretical advantages, the specific impacts of V-angled perforated fins on the thermal performance and overall efficacy of DPSACs still need to be comprehensively investigated in the existing literature. This study addresses this research gap by conducting experimental investigations to assess the effects of V-angled perforated fins on the efficiency of DPSACs.

The main objective of this research is to evaluate the influence of novel V-angled perforated fins on the thermal efficiency of DPSACs. The study examines critical performance metrics, such as thermal efficiency and temperature distributions, along with Nusselt and Reynolds numbers in the upper channel. The research involves a comparative analysis of the performance between a fin-equipped DPSAC and a baseline model without fins under three different flow rates. By providing empirical data on the performance enhancements offered by these fins, this study aims to inform future advancements in solar collector design and contribute to the development of sustainable energy solutions for heating applications.

2. Methodology

2.1. Setup details

Two Double-Pass Solar Air Collectors (DPSACs) employing forced convection were constructed for this experimental study. The study was conducted at Solar Lab, MATE university, Gödöllő, Hungary (47.59 N, 19.36 E). Each collector featured a wooden frame for the structure and a copper absorber plate (121 cm × 46 cm × 1.2 mm) for optimal heat transfer. A transparent plastic sheet was used to cover the collector, ensuring that the maximum amount of solar radiation reached the absorber. The first airflow path includes a 5.5 cm air passage formed between the absorber and the cover. Underneath the absorber plate, a second air passage measuring 35 mm was established. Six wooden buffers were strategically positioned to guide the airflow, enhancing distribution beneath the absorbing surface. The detailed design configurations of the collectors are depicted in Fig. 1.

The key focus of this study was the influence of V-angled perforated fins on thermal performance of the DPSACs. These fins were integrated into one of the collectors and are illustrated in Fig. 2. Each V-shaped fin, with a 90° angle between its 2.5 cm sides, was placed at a 45° angle relative to the absorber plate, creating an effective angle of attack for the airflow. Each side of the fin contained 78 systematically distributed holes (39 holes of 8 mm diameter and 39 holes of 6 mm diameter) along its 45 cm length, with no directly opposing holes of the same size. The fins were spaced 100 mm apart, with nine fins across the absorber plate,

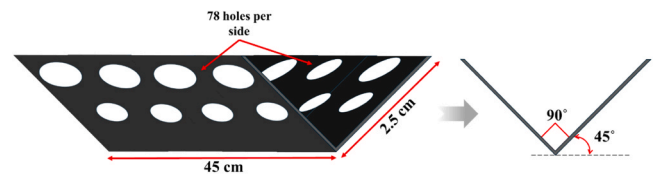


Fig. 2. 3D section illustration of V-angled perforated fins.

and secured using bolts and high-quality thermal silicone adhesive for optimal thermal contact.

This detailed design facilitated a comprehensive comparative analysis of the thermal efficiency between the two DPSACs, particularly examining the influence of V-angled perforated fins on heat transfer and airflow dynamics under real-world weather conditions. A variable-speed fan was integrated to achieve precise airflow management within the collector system, allowing for accurate airflow rate regulation through a fan dimmer. All collector components were accurately assembled, tightly secured, and sealed with premium acrylic sealant to minimize air leakage. The collectors were mounted on a south-facing aluminium frame tilted at a 45° angle to maximize sunlight capture. This careful design and alignment underscore the study's dedication to optimizing solar power utilization.

2.2. Measurements devices

Temperature is a crucial parameter in this study, for which thirty T-type thermocouples were strategically placed. Four thermocouples were used for the inlet and another four for the outlet, positioned 150 mm from the ports, aiming to maintain temperature variations within 0.5°C. Additionally, six thermocouples were situated on each absorber plate, and two more measured the ambient temperatures, with readings averaged for accuracy. The thermocouples were calibrated and tested before installation. Relative humidity was monitored using humidity sensors. The airflow rates were measured using an orifice plate meter that functions based on Bernoulli's principles. Analogue differential pressure sensors measured the pressure difference across the orifice plate and the collector channels. Solar radiation was recorded by means of pyranometer. The data from these sensors were logged using an Advantech ADAM system. The description of the measurement instruments, along with their accuracy and function, is presented in Table 1. Fig. 3 illustrates a schematic view, while Fig. 4 provides a graphical representation of the device's setup.

2.3. Experimental procedure

The experiments aimed to evaluate the performance of smooth (DPSAC) and V-angled finned (FDPSAC) double-pass solar air collectors under outdoor test conditions. Both collectors were tested side-by-side to assess the impact of their configurations on performance solely. Before the experiments, great attention was given to ensure the collectors were correctly assembled to the flow pipes, securely fastened, and sealed to significantly minimize the risk of air leakage. Additionally, all temperature and pressure sensors were calibrated and checked at steady state, showing almost identical readings for both collectors. The experiments took place on mostly clear days between August 15th and August 22nd, 2023.

To achieve the objectives of the study three distinct airflow rates were selected for testing both collectors: 0.00864 kg/s (case I on August 15th), 0.01143 kg/s (case II on August 21st), and 0.01317 kg/s (case III on August 22nd).

The following data were collected through the experiments every minute from 10:30–15:30 and graphed at five-minute intervals:

- The ambient temperature average of two sensors.
- The inlet temperature average of four sensors.

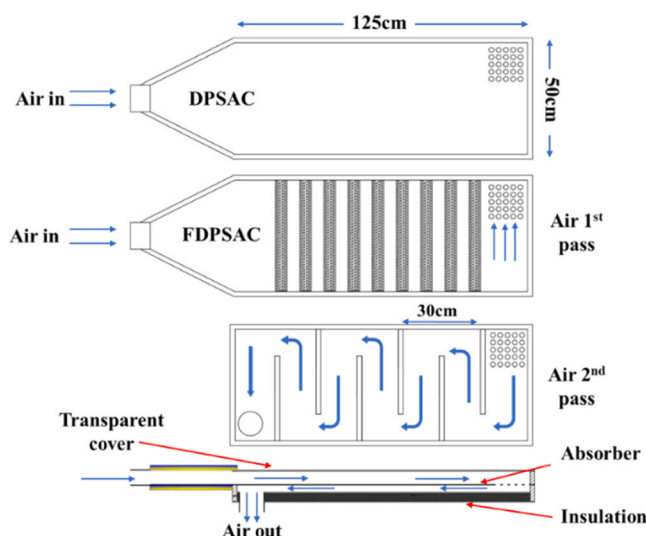


Fig. 1. Shows the structure layout of the collectors.

Table 1
Specifications and accuracy of measurement instruments.

No	Instrument	Model	Range	Accuracy	Function
1	Temperature Thermocouples	T-type	-250–250C	±0.5C	temperature measurements
2	Relative Humidity Sensor	Honeywell HIH-4000-004	0–100 %	±3.5 %	air humidity sensor
3	Differential pressure sensor	SENSIRION SDP-810	-500–500 Pa	±3 %	measure pressure drops through the collector and the orifice plate
4	Solar Radiation Pyranometer	Kipp & Zonen CM11	0–1400 W/m ²	<0.6 % for readings below 1000 W/m ²	measures solar radiation intensity
5	Data Logger (Temperature)	Advantech ADAM 4018, 4018+, 4019+	-100–400C	±0.1 %	logs temperature data from multiple locations
6	Data Logger (Pressure, Humidity, Radiation)	Advantech ADAM 4017	0–10 V	±0.1 %	logs pressure, humidity, and solar radiation data

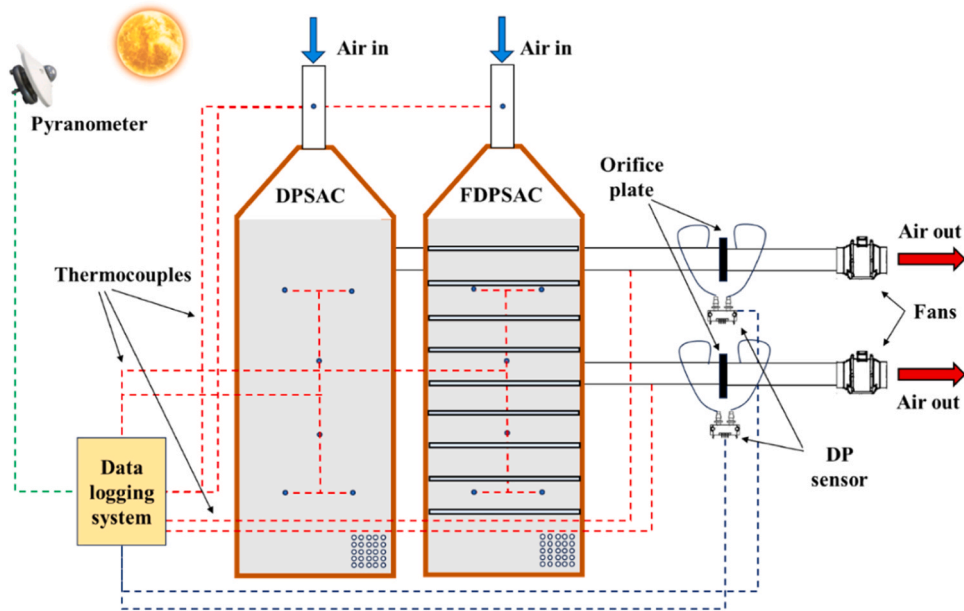


Fig. 3. Schematic diagram of the experimental configuration.

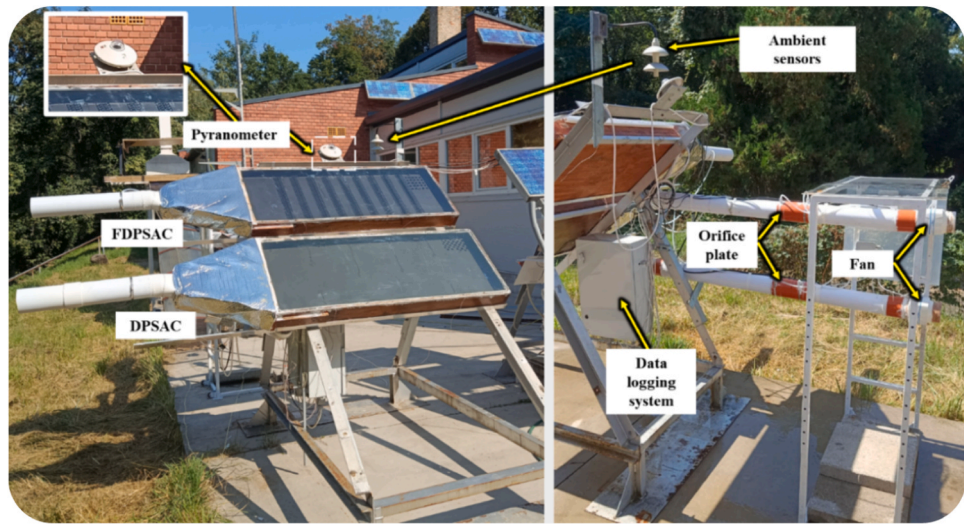


Fig. 4. Graphical illustration of the experimental arrangement.

- The absorber plate temperature average of six sensors.
- The outlet temperature average of four sensors.
- The orifice plate pressure drops ΔP_{or} .
- The pressure drops ΔP across the collector channel
- The solar radiation intensity

2.4. Theoretical analysis

Figs. 5 and 6 illustrate the thermal network of DPSACs with and without the presence of fins. In this context, the temperatures of the glazing cover, air in the first pass, absorbing plate, air in the second pass, and insulation are denoted as (T_g) , (T_{f1}) , (T_p) , (T_{f2}) , and (T_b) , respectively. Heat balance equations for each temperature point with assumptions critical for model simplification are derived as documented by Fudholi et al. (2011); Naphon (2005); Ong (1995).

Energy balance equations for DPSAC without fins, as outlined by the thermal network in Fig. 5, are introduced as follows:

Glass cover energy balance at T_g ,

$$I \alpha_g + h_{p-g}(T_p - T_g) + h_{f1-g}(T_{f1} - T_g) = U_t(T_g - T_a). \quad (1)$$

For the flowing air at the first pass at T_{f1} is,

$$h_{p-f1}(T_p - T_{f1}) = h_{f1-g}(T_{f1} - T_g) + Q_{u1}. \quad (2)$$

For the absorber plate at T_p ,

$$I \alpha_p \tau_g = h_{p-f1}(T_p - T_{f1}) + h_{p-f2}(T_p - T_{f2}) + h_{p-g}(T_p - T_g) + h_{p-b}(T_p - T_b). \quad (3)$$

For the flowing air at the second pass at T_{f2} ,

$$h_{p-f2}(T_p - T_{f2}) = h_{f2-b}(T_{f2} - T_b) + Q_{u2}. \quad (4)$$

For the bottom insulation surface at T_b ,

$$h_{f2-b}(T_{f2} - T_b) + h_{p-b}(T_p - T_b) = U_b(T_p - T_a). \quad (5)$$

Fig. 6 presents the thermal network outlines the energy balance equations for a DPSAC utilizing fins, showcasing how the presence of fins modifies the system to improve heat transfer and energy efficiency. The equations are presented as follows:

Glass cover energy balance at T_g ,

$$I \alpha_g + h_{p-g}(T_p - T_g) + h_{f1-g}(T_{f1} - T_g) = U_t(T_g - T_a). \quad (6)$$

For the first pass air flow with fins at T_{f1} ,

$$h_{p-f1}(T_p - T_{f1}) = h_{f1-g}(T_{f1} - T_g) + Q_{u1} + \frac{N}{A_c} Q_n. \quad (7)$$

For the absorber plate with fins at T_p ,

$$I \alpha_p \tau_g = h_{p-f1}(T_p - T_{f1}) + h_{p-f2}(T_p - T_{f2}) + h_{p-g}(T_p - T_g) + h_{p-b}(T_p - T_b) + \frac{N}{A_c} Q_n. \quad (8)$$

Second pass air flow energy balance at T_{f2} is,

$$h_{p-f2}(T_p - T_{f2}) = h_{f2-b}(T_{f2} - T_b) + Q_{u2}. \quad (9)$$

Bottom insulation surface energy balance at T_b ,

$$h_{f2-b}(T_{f2} - T_b) + h_{p-b}(T_p - T_b) = U_b(T_p - T_a). \quad (10)$$

Where Q_{u1} , Q_{u2} , Q_{fin} represent useful energy from the first pass, second pass useful energy, and the fin heat transfer rate, respectively, these variables attain their final form after simplification and integration as follows Fudholi et al. (2011); Ong (1995):

$$Q_{u1} = 2\dot{m}C_p(T_{f1} - T_{f1i})/WL, \quad (11)$$

$$Q_{u2} = 2\dot{m}C_p(T_{f2} - T_{f2i})/WL, \quad (12)$$

$$Q_n = (2k_p A_{fin} h_{fin})^{1/2} (T_p - T_{f1}) \tanh KH. \quad (13)$$

In this context, h_{fin} stands for the heat transfer coefficient, k_p denotes thermal conductivity, A_{fin} signifies fin area, H fin height, and l indicates fin length. The value of K is determined through the following equation Fudholi et al. (2011):

$$K = \left(\frac{2lh_{fin}}{k_p A_{fin}} \right)^{1/2}. \quad (14)$$

Ong (1995) outline the methodology for calculating the heat transfer coefficients due to radiation, identified as h_{rg-s} , h_{rp-g} , and h_{rp-b} , as follows:

For the top plastic cover to sky h_{rg-s} :

$$h_{rg-s} = \frac{\sigma \epsilon_g (T_g + T_s) (T_g^2 + T_s^2) (T_g - T_s)}{(T_g - T_a)}. \quad (15)$$

From the absorbing plate to plastic cover h_{rp-g} :

$$h_{rp-g} = \frac{\sigma (T_g^2 + T_p^2) (T_g + T_p)}{\left(\frac{1}{\epsilon_g} + \frac{1}{\epsilon_p} - 1 \right)}. \quad (16)$$

For the absorber plate to bottom insulation surface h_{rp-b} :

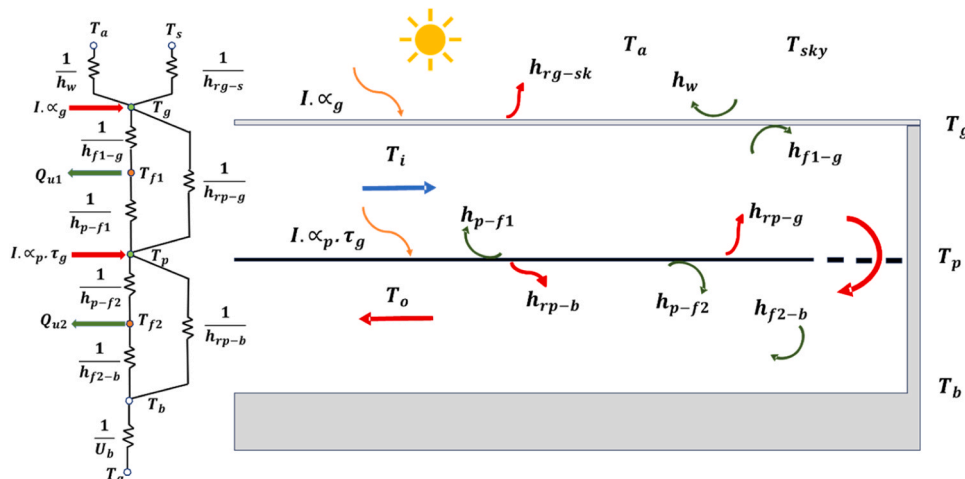


Fig. 5. Illustrate the thermal network of DPSAC.

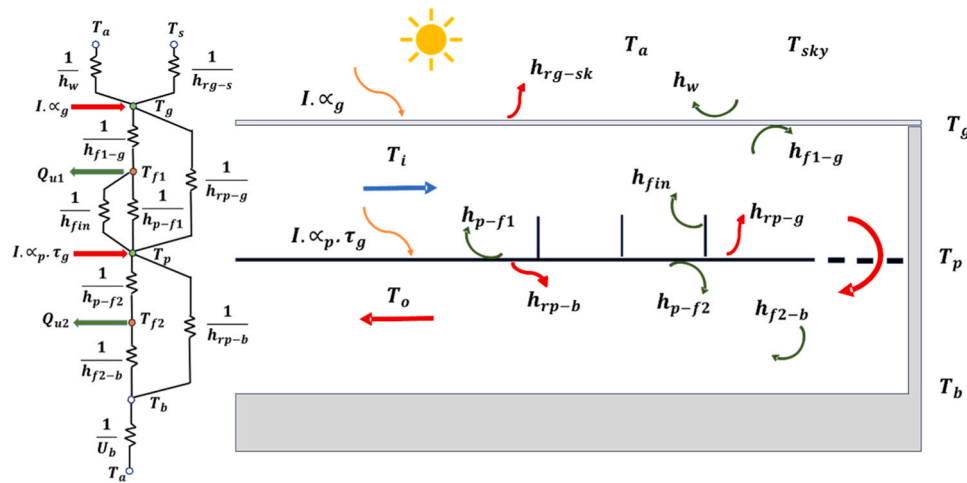


Fig. 6. Demonstrate the thermal network of FDPSAC.

$$h_{p-b} = \frac{\sigma(T_p^2 + T_b^2)(T_p + T_b)}{\left(\frac{1}{\varepsilon_p} + \frac{1}{\varepsilon_b} - 1\right)}. \quad (17)$$

Where σ denotes the Stefan-Boltzmann constant, and ε_g , ε_p , and ε_b are the emissivity of the plastic cover, absorber plate, and bottom insulation, respectively. Sky temperature T_s , linked to ambient temperature T_a , is determined by Eq. 18. Convective heat transfers due to wind, h_w dependent on wind speed V_w , is calculated by Eq. 19. The overall top heat loss coefficient U_t is the sum of h_w and h_{rg-s} , as Eq. 20. Heat loss from the back cover, U_b , through conduction is also considered as shown in Eq. 21 Assadeg et al. (2021); Ong (1995).

$$T_s = 0.0552T_a^{1.5}. \quad (18)$$

$$h_w = 2.8 + 3.3V_w. \quad (19)$$

$$U_t = h_w + h_{rg-s}. \quad (20)$$

$$U_b = \frac{1}{\sum_{i=1}^n (x_{bi}/k_{bi}) + 1} / h_w. \quad (21)$$

All convective heat transfer coefficients within the collector flow channels are determined according to the equation provided by Assadeg et al. (2021); Ong, (1995).

$$h_{p-f1} = h_{f1-g} = h_{f2-b} = h_{p-f2} = h_{fin} = Nu_{D_h} \frac{k}{D_h}. \quad (22)$$

In this context, Nu indicates the Nusselt Number, k is the air thermal conductivity, while D_h is the hydraulic diameter of the ducts, and. For laminar flow region with Reynolds (Re) ($Re < 2300$), the transition flow region where ($2300 < Re < 6000$), and the turbulent flow region with ($Re > 6000$), formulas for Nusselt and Reynolds numbers follow the guidelines provided in references Naphon (2005); Ong (1995).

2.5. Data reduction

The instantaneous efficiency of SAC can be calculated by the following expression Kabeel et al. (2016):

$$\eta = \frac{Q_u}{IA_p}. \quad (23)$$

Where A_p denotes the projected area of the collector (m^2), I represent the solar radiation incident on the collector (W/m^2), and Q_u is the useful energy of the air through the collector, which is defined as follows:

$$Q_u = \dot{m} C_p \Delta T. \quad (24)$$

Hence $\dot{m}$ denotes the airflow rate (kg/s) calculated using Eq. 25 Alam et al. (2014a); Singh et al. (2012), C_p represents the air specific heat (J/kg°C), and ΔT is the temperature difference of the air across the collector.

$$\dot{m} = C_d \times A_{or} \times \left[\frac{2 \times \rho_o \times \Delta P_{or}}{1 - \beta^4} \right]^{0.5}. \quad (25)$$

Where, C_d indicates the discharge coefficient of the orifice plate, while A_{or} is the area of the orifice. The air density at the outlet is represented by ρ_o , and ΔP_{or} refers to the pressure difference across the orifice. Diameter ratio of the orifice to the duct is denoted by β .

The Reynolds number (Re) for the airflow through the test section is calculated using the following correlation:

$$Re = \frac{\dot{m} D_h}{A_{ch} \mu}. \quad (26)$$

Where the hydraulic diameter of the ducts D_h estimated as following:

$$D_h = \frac{4WB}{2(W+B)}. \quad (27)$$

The convective heat transfer coefficient h estimated by below correlation:

$$h = \frac{Q_u}{A_p(T_p - T_f)}. \quad (28)$$

The friction factor (f) computed from the pressure drop across the collector passes:

$$f = \frac{2\Delta p D_h}{4\rho L V^2}. \quad (29)$$

2.6. The analysis of the uncertainty

Experimental data gathered in research can differ from actual values due to uncontrolled experiment factors. This variation between observed and actual data is termed uncertainty. The procedure followed in this study is documented in previous publications by Kumar (2019); Omojaro and Aldabbagh (2010).

The general equation used is Eq. 30:

$$\delta_R = \left[\left(\frac{\delta R}{\delta x_1} \delta x_1 \right)^2 + \left(\frac{\delta R}{\delta x_2} \delta x_2 \right)^2 + \left(\frac{\delta R}{\delta x_3} \delta x_3 \right)^2 + \dots + \left(\frac{\delta R}{\delta x_n} \delta x_n \right)^2 \right]^{0.5} \quad (30)$$

Variables x_1, x_2, x_3, x_n may contain errors, with δx_n indicating potential error, δR representing absolute uncertainty. The relative uncertainty of the investigated parameters presented in Table 2.

3. Results and discussion

This section presents a detailed comparison between finned and smooth DPSACs. The results involve several key parameters: plate temperature, outlet temperature, temperature difference, useful energy, thermal efficiency, Nusselt number, friction factor due to fins. By examining these aspects, we aim to understand how fins affect heat transfer and airflow resistance. This comparison helps identify the optimal design for achieving high efficiency with minimal drawbacks in DPSAC systems.

Fig. 7 illustrates the changes in surrounding ambient temperature and relative humidity during the experiments. The ambient air temperatures for cases II and III were relatively closer, but case I exhibited a slight difference. Nonetheless, the ambient temperature for all cases varied between 28.5°C to 35.5°C. On the other hand, the relative humidity fluctuated between 46.6 % and 61.3 % for all cases, with case III showing a notably higher percentage. Minor differences between testing days are linked to Gödöllő's climatic conditions. These fluctuations may affect the overall efficiency of solar air collectors. However, they do not alter the comparative evaluation of the two collectors, as they were assessed concurrently under the same conditions. Therefore, the effect of environmental conditions during the experimental period can be dismissed.

Fig. 8 illustrates the solar intensity recorded during the investigational days. The solar radiation rises from the morning, peaking at noon, and declines towards the experiment's end at 15:30. After this time, despite the availability of sunlight, the declining radiation levels, dropping below 500 W/m², and shading render further measurements ineffective for our purposes. The maximum solar radiation observed for case I was 914.8 W/m², with cases II and III reaching 925.6 W/m² and 934.3 W/m², respectively.

Fig. 9 illustrates the absorber plate temperature for both collectors in each test case over time. The efficiency of SAC is greatly impacted by the absorber plate temperature, affecting it positively and negatively. A positive influence occurs when a sufficient temperature difference exists between the plate and the circulating fluid, enhancing the heat transfer rate. Conversely, a negative impact arises when the temperature becomes excessively high, leading to increased heat losses between the collector and the ambient environment, as Rani and Tripathy (2020) noted. As illustrated, absorber plate temperatures rise in the morning with increasing solar radiation intensity, peaking around solar noon for each case. Subsequently, temperatures fall as solar intensity decreases. Additionally, the pattern observed in the absorber plate temperature closely reflects the solar radiation intensity, as demonstrated in Fig. 8. This indicates a significant influence of solar power on the absorber temperature. Moreover, plate temperature is inversely related to airflow rate, decreasing as mass flow rate increases, and increasing as mass flow rate decreases. Remarkably, the plate temperature of FDPSAC remains

Table 2

Measurement uncertainties for investigated parameters.

Parameter	Relative uncertainty
Airflow rate	1.55 %
Thermal efficiency	2.36 %
Reynolds number	1.59 %
Nusselt number	3.39 %
Friction factor	3.42 %

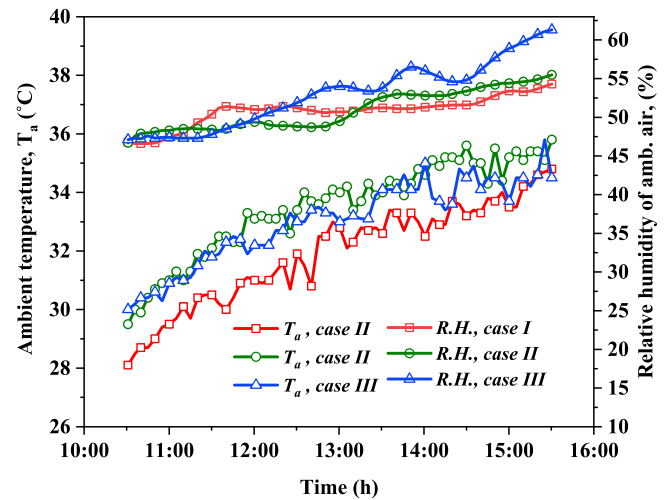


Fig. 7. The surrounding ambient temperature and its relative humidity versus time.

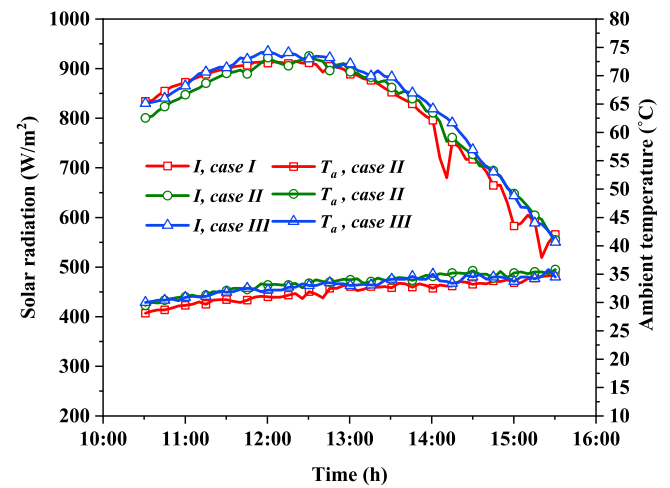


Fig. 8. The solar radiation and the surrounding temperature versus time.

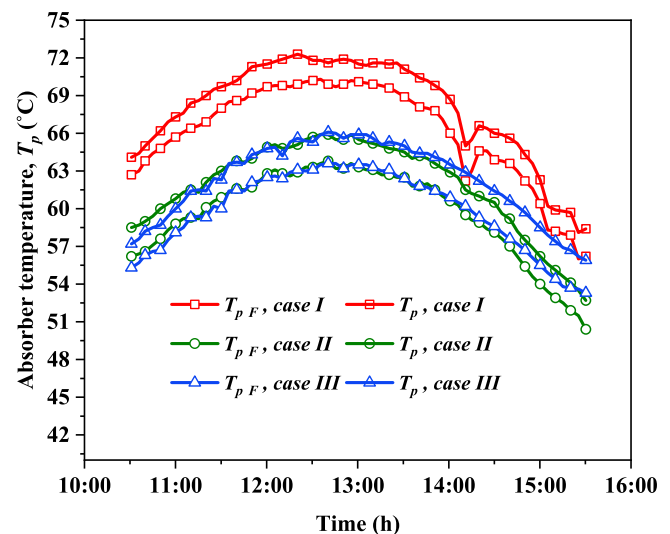


Fig. 9. The absorbing plate temperature profiles versus time.

lesser of the DPSAC, attributable to the fins' role in augmenting the heat transfer rate, hence reducing absorber temperature. This finding aligns with previous research conducted by Alomar et al. (2022); Hassan and Abo-Elfadl (2018) highlighting the fins' effectiveness in enhancing thermal performance.

Similarly, the connection between solar intensity and plate temperature is reflected in the outlet temperatures of the collectors, demonstrating a comparable pattern as shown in Fig. 10. For both collectors, as solar radiation intensity rises throughout the morning, the outlet temperature (T_o) increases until reaching a peak around noon. Subsequently, there is an obvious decline in temperature, observed gradually until the conclusion of the experiment at 15:30. Additionally, the figure shows that T_o is influenced by both the flowrate and the radiation intensity. An inverse relationship is observed, where T_o decreases as the airflow rate increases for both collectors. The maximum recorded outlet temperatures for the FDPSAC collector were 60.6°C, 57°C, and 56.2°C for cases I, II, and III, respectively, between 12:30 and 13:30. Similarly, the DPSAC collector reached highest temperatures of 59°C, 55°C, and 53°C for the similar cases. Notably, T_o of the FDPSAC consistently remains higher than that of the DPSAC under all airflow rate conditions. The improvement in the rate of heat transfer, attributed to the V-angled fins, stems from an expanded surface area and heightened airflow turbulence, disrupting the stagnant layer between the air in motion and the absorbing surface. This observation aligns with the findings reported by Abd et al. (2022); Alam and Kim (2017); Vengadesan et al. (2024), underscoring the thermal benefits of such fin configurations.

Fig. 11 illustrates the variation relating inlet and outlet temperatures ($\Delta T = T_o - T_i$) of the tested collectors over time. The collectors, tested outdoors, experienced inlet air temperatures at or below ambient levels due to forced airflow. The profile of temperature differences aligns with trends observed in Figs. 9 and 10, highlighting the substantial impact of sun radiation and ambient temperature upon these variations. Additionally, mass flow rate through the collectors directly impacts the temperature difference, with higher airflow rates resulting in lower temperature differentials, so it crucial to be balanced. Moreover, the FDPSAC consistently showed greater temperature differences compared to the DPSAC at all airflow rates. Peak temperature differentials observed for FDPSAC were 30.7°C, 25.8°C, and 24.1°C from 12:30–13:30 for cases I, II, and III, respectively, exceeding those recorded of DPSAC, which were 29.0°C, 23.8°C, and 21.5°C. These findings highlight the vital role of fins in improving performance by enhancing heat transfer area as well as optimizing flow behaviour. The results align with earlier research findings reported in Abd et al. (2022); Abdullah et al. (2018); Omojaro and Aldabbagh, (2010); Yassien et al. (2020).

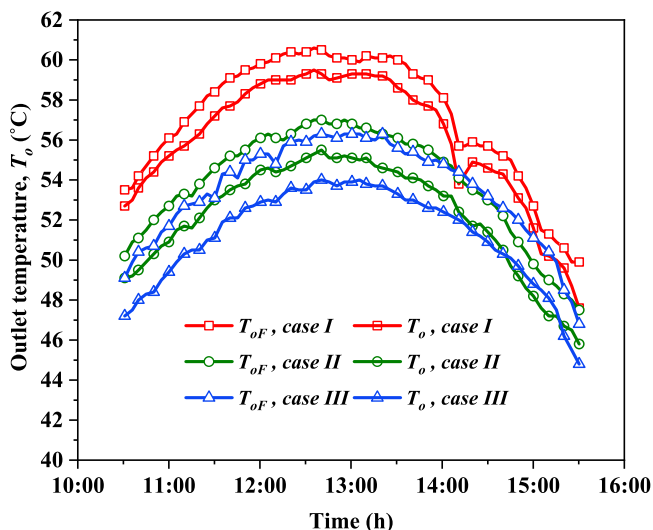


Fig. 10. Outlet temperature profiles for the collectors over time.

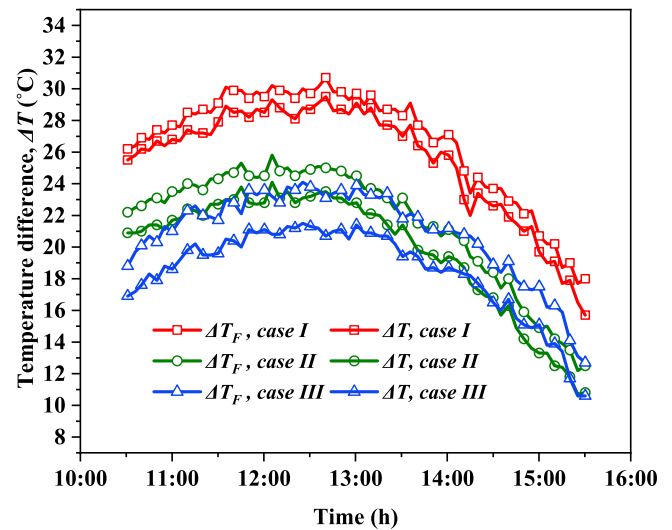


Fig. 11. The temperature difference across the collectors versus time.

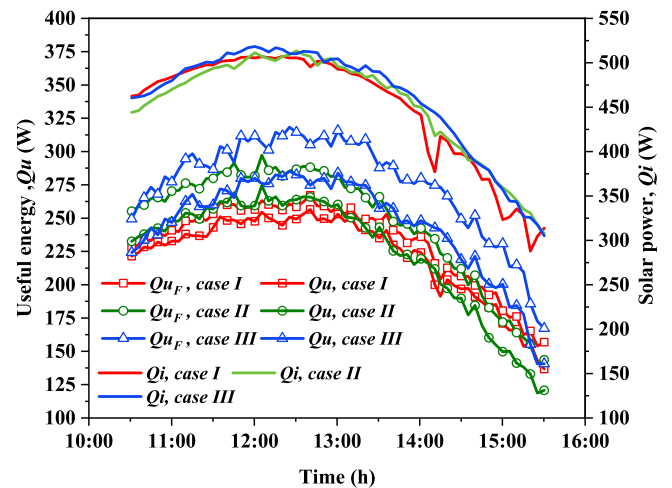


Fig. 12. The energy gained by the collectors over time.

Fig. 12 demonstrates the profile of the energy gained over time for the collectors. As observed in the figure, the energy gained raises with time from middy morning till reaching its maximum around noon and then declining till the end of the experiments, after which the energy is minimal. Moreover, the energy profile trend follows the solar power energy curve ($Q_i = I \times A_p$) for all cases, as presented in the figure. This alignment indicates that the energy gained is constrained by the flow rate and the temperature variation ΔT . The limitation is due to the fact that temperature variation is governed by the outlet temperature, which, in turn, is impacted by the radiation intensity. Furthermore, temperature difference is a key factor in the process of heat gain, representing the most significant variable in the calculation equation. Additionally, airflow rate directly influences useful energy, with higher mass flow rates enhancing heat transfer. The FDPSAC consistently outperforms the DPSAC in energy gain across all cases due to improved heat transfer from fins, which expand the heat transfer area and disrupt the boundary layer, enhancing efficiency. The maximum heat energy gains for FDPSAC, recorded between 12:00 and 13:00, were 267.1 W, 297.3 W, and 318.3 W corresponding to cases I, II, and III, respectively, while for the DPSAC were 256.5 W, 274.4 W, and 289 W for the same cases. The above results align with findings in literature as reported in Alomar et al. (2022); Kabeel et al. (2016); Machi et al. (2024)

Efficiency curves for solar collectors typically rise and fall from

sunrise to sunset through the day, relatively mirroring the solar radiation curve. This pattern is expected when experiments span from early morning to sunset. However, in practice, conducting tests at low radiation levels can be inefficient and costly due to the energy expended on operating the air circulation pump. Therefore, in these experiments, data collection began at 10:30, when solar radiation exceeded 750 W/m² and no shading impacted the collector, concluding at 15:30 as radiation levels fell below 500 W/m², rendering further testing ineffective.

Fig. 13 corresponds to these expectations, displaying an initial increase in efficiency, followed by fluctuations until peak values are reached around noon; subsequently, they gradually decline until the end of the test. The collector's instantaneous efficiency directly correlates to airflow rates due to enhanced heat transfer rates, which augment heat gain and optimize efficiency. Despite variations in airflow rates, the FDPSAC consistently demonstrates superior instantaneous efficiency linked to its fin design that improved heat energy transfer and consequently elevating overall efficiency. The maximum calculated efficiency for FDPSAC was 58.2 %, 59.4 %, and 64.8 % corresponding to cases I, II, and III, respectively, while DPSAC achieved 54.7 %, 54.8 %, and 57 % for the same cases. These findings align with previous research Abd et al. (2022); Kabeel et al. (2016); Ozgen et al. (2009); Salih et al. (2021). Additionally, the instantaneous efficiency is highly sensitive to solar radiation intensity. A drop in radiation intensity caused by a sudden cloud cover or an error in solar radiation data can initially result in higher efficiency. The reason is that the absorber still has stored heat, and the energy gain remains constant while the input energy decreases. Consequently, peak instantaneous efficiency may not accurately reflect the collector performance, making the calculation of daily thermal efficiency essential for an accurate evaluation.

As outlined, daily efficiencies were calculated utilizing the trapezoidal rule for numerical integration, ensuring precise performance assessment as described in Machi et al. (2022). The results in Fig. 14 show that the FDPSAC outperforms the DPSAC in terms of daily efficiency across all tested airflow rates. Notably, the FDPSAC achieved daily efficiency values of 51.6 %, 54.7 %, and 59.2 % at mass flow rates of 0.00864 kg/s, 0.01143 kg/s, and 0.01317 kg/s, respectively. In contrast, the DPSAC reported lower efficiencies of 49.4 %, 50.5 %, and 53.9 % for the same airflow rates. The data reveals that the innovative V-angled perforated fin design of the FDPSAC accounts for substantial relative efficiency improvements of approximately 4.45 %, 8.32 %, and 9.83 % compared to the DPSAC. These significant efficiency gains validate the effectiveness of the fin design in improving the solar collector's efficiency. The present study found that the newly proposed configuration exhibited superior efficiency, corroborating previous

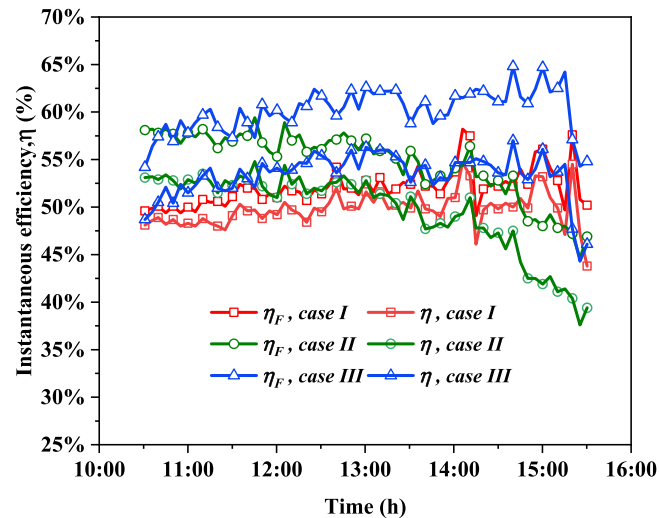


Fig. 13. Instantaneous thermal efficiency over time.

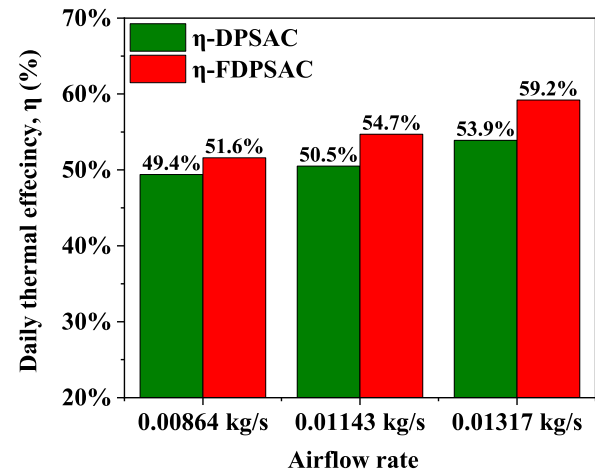


Fig. 14. The daily thermal efficiency versus airflow rate.

Table 3
Provides a comparative between the findings of the present study and those documented in Literature.

Reference	Study type	Collector type and configuration	Efficiency indicators
Al-Damook et al. (2019)	Numerical and experimental	DPSAC, Aluminium cans extended surfaces with two configurations in line and staggered.	The efficiency of staggered configuration reached about 60.2 % at MFR of 0.096 kg/s
Hussein et al. (2023)	Numerical and experimental	DPSAC with tubular absorber parallel and perpendicular to the flow direction.	The effective efficiency of the parallel configuration was 80.9 % at 0.03 kg/s
Sudhakar and Cheralathan, (2020)	Experimental	DPSAC has V-groove absorber plate with pin fins in the upper and lower channel.	The maximum efficiency achieved was 52 % at 0.035 kg/s with average increase of 17.4 %
Salih et al. (2019)	Experimental	Trapezoidal DPSAC with six rectangular fins	The maximum efficiency achieved 67 %, at 0.032 kg/s
El-Sebaei et al. (2011)	Numerical and experimental	DPSAC, v-corrugated plate	The maximum efficiency achieved 65.3 % at 0.04 kg/s with increase of 11–14 %
El-khawajah et al. (2011)	Experimental	DPSAC, transverse fins with wire mesh as an absorber	The maximum efficiency reached 85.9 %, at 0.032 kg/s
Farzan and Zaim, (2023)	Experimental	DPSAC, corrugated and perforated absorber with porous medium	The daily efficiency achieved 66.1 % at 0.024 kg/s relative increase of 15.3 %
Present work	Experimental	DPSAC, V-angled perforated fins in the first pass	The daily efficiency exceeded 59 % at 0.01317 kg/s with relative increase of 9.83 %

research findings, as shown in Table 3. When considering a 0.01317 kg/s flow rate, the proposed collector showed improved performance compared to those documented in the existing literature.

Figs. 6 and 7 illustrate the variations in the Nusselt number and friction factor with the Reynolds number in the upper path due to fins presence, respectively. As *Re* rises for both types of collectors, the *Nu* also rises, whereas the friction factor declines. This trend is consistent

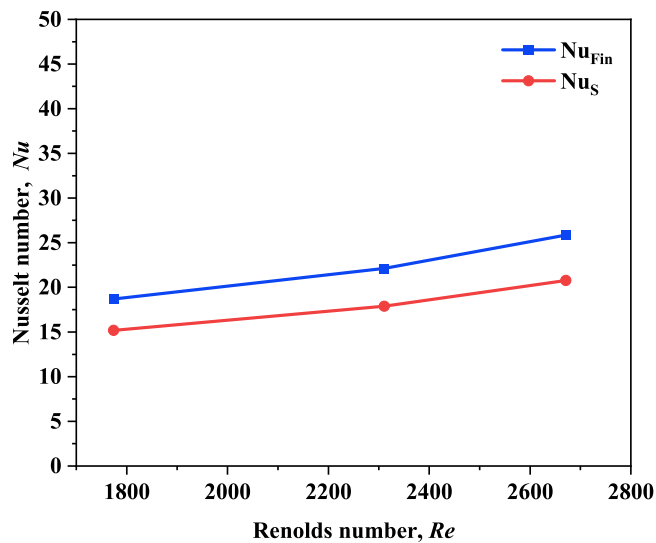


Fig. 15. The Nusselt number versus Renolds number.

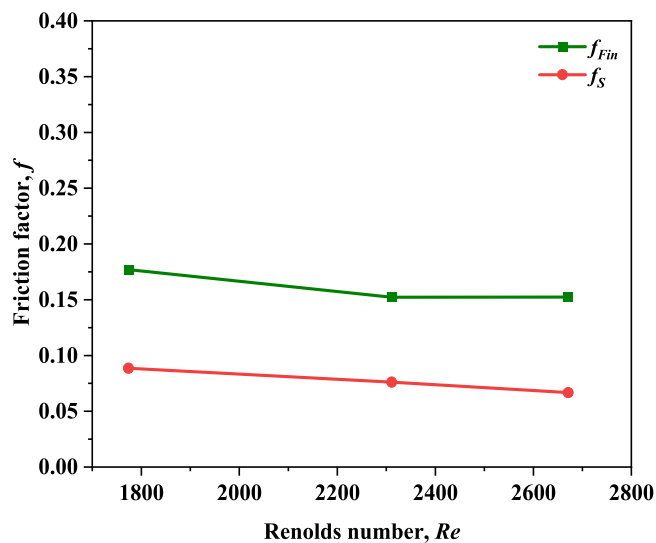


Fig. 16. The friction factor versus Renolds number.

with theoretical expectations, owing to the enhanced turbulence. The finned collector exhibits a higher Nu compared to the smooth collector, with an average relative improvement between 23 % and 24.4 %. However, the friction factor significantly increased, reaching up to 126 % due to the fins. Although the proposed perforated fins cause only a minor change in pressure drop compared to the smooth plate type, they still lead to a significant increase in the friction factor. Therefore, it is important to consider this effect when choosing fin designs. The findings highlight the importance of balancing the improved heat transfer with the additional pumping power required due to the increased friction. The results align with findings in literature as reported in Alam et al. (2015); Hedau and Singal, (2023); Sharma and Debbarma, (2024)

4. Conclusion

This study investigates the impact of innovative V-angled perforated fins on the efficiency of double-pass solar air collectors (DPSAC) across various flow rates. A comparative study was performed between a finned double-pass solar air collector (FDPSAC) with a novel fin and a standard DPSAC lacking these modifications. Both systems were

evaluated under identical conditions at three different airflow rates: 0.00864 kg/s (case I), 0.01143 kg/s (case II), and 0.01317 kg/s (case III). The findings underscore the significant contribution of V-angled perforated fins in enhancing collector efficiency. Key results are as follows:

- The FDPSAC consistently achieved higher outlet temperatures across all tested airflow rates compared to the DPSAC. Specifically, the outlet temperatures of the FDPSAC reached 60.6C, 57C, and 56.2C for cases I, II, and III, respectively, while the DPSAC recorded lower temperatures of 59C, 55C, and 53C for the corresponding cases. This indicates the fins' effectiveness in enhancing heat transfer.
- A significant finding was the inverse relationship between airflow rate and temperature difference for both collectors, emphasizing the critical role of airflow regulation in optimizing DPSAC efficiency. The temperature differences ΔT were consistently higher in the FDPSAC, demonstrating its superior capacity to utilize absorbed heat.
- Absorber plate temperatures were significantly lower in the FDPSAC across all scenarios, confirming the fins' role in promoting efficient heat dissipation. This highlights the fins' ability to moderate excessive heat accumulation, thereby reducing thermal losses and enhancing overall performance.
- The efficiency of the standard DPSAC was lower compared to the FDPSAC. The integration of V-angled perforated fins in the FDPSAC facilitates improved airflow distribution within the collector, effectively addressing issues related to stagnant zones. The observed performance enhancements in the FDPSAC indicate that incorporating V-angled perforated fins significantly improves heat transfer rates.
- The daily efficiencies for the FDPSAC were 51.6 %, 54.7 %, and 59.2 % across cases I, II, and III, respectively. In contrast, the DPSAC exhibited efficiencies of 49.4 %, 50.5 %, and 53.9 % for the identical case order. The FDPSAC showed relative efficiency improvements of approximately 4.45 %, 8.32 %, and 9.83 % compared to the DPSAC. These efficiency gains highlight the potential of such innovations to advance solar collector technologies.

Based on these findings, the study recommends the following areas for future exploration to expand the knowledge and utilization of fin integrated DPSAC designs:

1. Explore the optimization of fin configurations, focusing on angles, perforation patterns, and spacing to maximize thermal performance and airflow dynamics within DPSACs.
2. Develop sophisticated computational simulations to model the airflow dynamics and efficiency of DPSACs featuring different fin configurations, enabling a more comprehensive analysis of the underlying processes.
3. Investigate how various environmental factors, such as changes in solar radiation intensity, ambient temperature, and humidity levels, affect the performance of DPSACs with fins.

CRedit authorship contribution statement

Maytham H. Machi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **István Farkas:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **János Buzas:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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